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Lab Practical #03:

Study of different types of network cables & connectors and crimping a LAN.

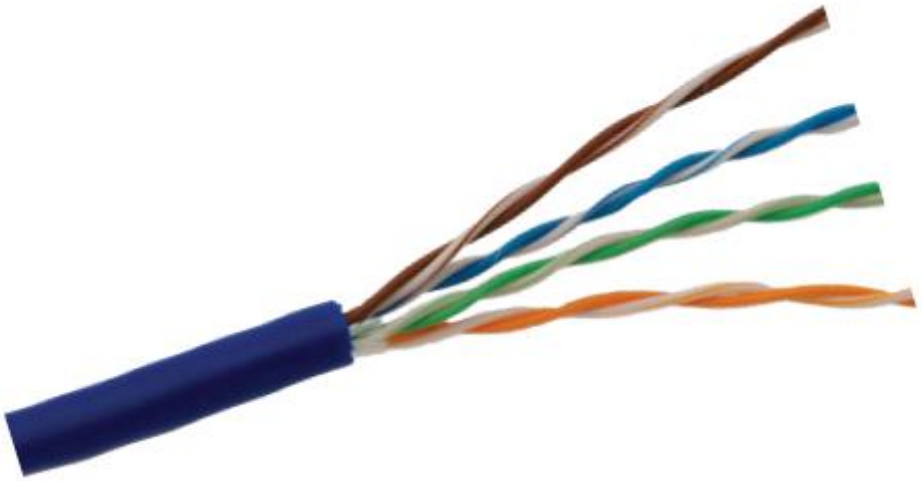
Practical Assignment #02:

1. List various networks cable. Also, write short details and usage.
2. Difference between guided and unguided media.
3. Give cross-wired cable and straight through cable diagram (Color Code wise).

1. List various networks cable and connectors. Also, write short description.

a) Twisted Pair Cable:

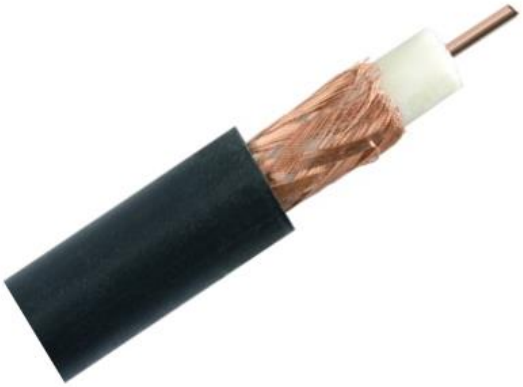
- **Description**:
  Used in LANs. UTP (Unshielded Twisted Pair) is common in homes/offices. STP (Shielded Twisted Pair) includes shielding to prevent interference.
- **Diagram**:



b) Coaxial Cable:

- **Description:**
  Contains a core conductor surrounded by insulation, shielding, and outer jacket. Used in cable TV and early Ethernet.
- **Diagram:**

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c) **Fiber Optic Cable:**
   - **Description**:
     Transmits data as light. Offers very high speed and long-distance transmission. Immune to electromagnetic interference.
   - **Diagram**:



2. Difference between guided and unguided media.

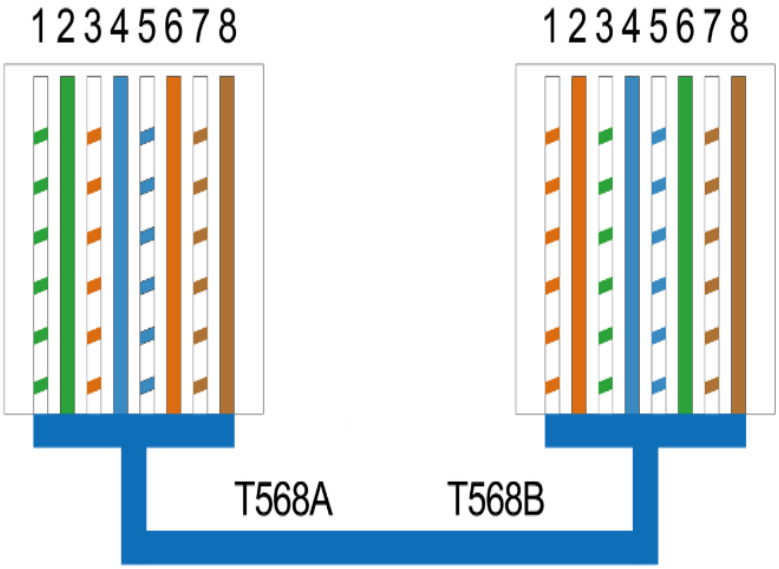
No.	Guided Media	Unguided Media
1	Uses physical path (cables) to transmit signals.	Transmits signals wirelessly through air.
2	Examples: Twisted Pair, Coaxial, Fiber Optic.	Examples: Radio waves, Microwaves, Infrared.

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3	High security and less susceptible to interference.	More vulnerable to interference and signal loss.
4	Installation cost is higher.	Cheaper and easier to deploy over large areas.
5	Directional; point-to-point or point-to-multipoint.	Mostly omnidirectional broadcast.

3. Give cross-wired cable and straight through cable diagram & usage (Color Code wise).

a) Cross-wired Cable Diagram (Color Code)



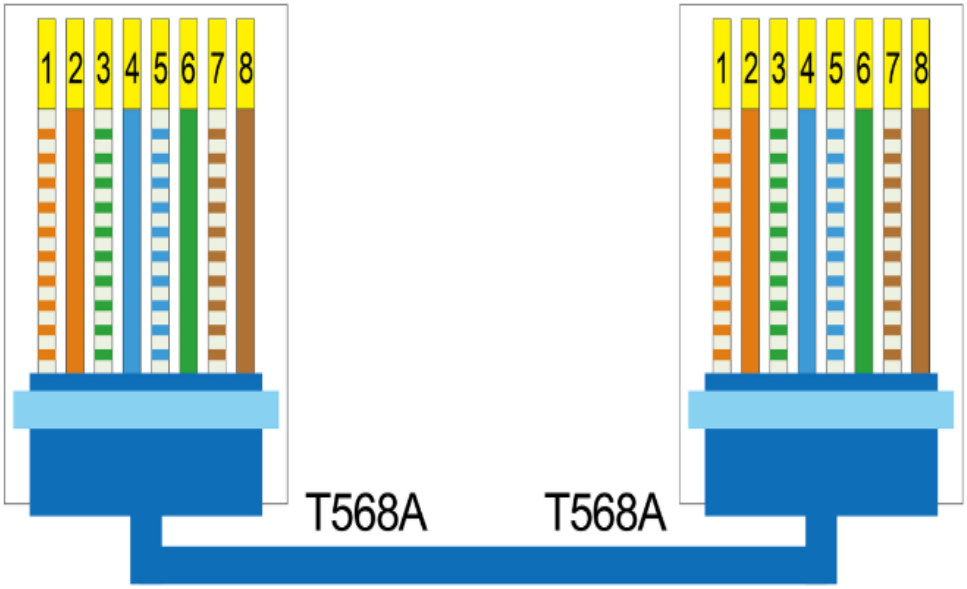
A straight-through cable is designed to connect two different types of networking equipment. Because the devices operate at different layers or have different default roles, one device naturally transmits data on the exact pins where the other device expects to receive it.

Common Use Cases:

- Computer (NIC) to a Switch or Hub
- Router to a Switch or Hub
- Modem to a Router or Computer

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b) Straight Through Cable Diagram (Color Code)



A crossover cable is designed to connect two identical or highly similar devices without a middleman (like a switch or hub).

Common Use Cases:

- Computer to Computer directly (e.g., for direct file sharing)
- Switch to Switch (to expand network ports)
- Hub to Hub
- Router to Router